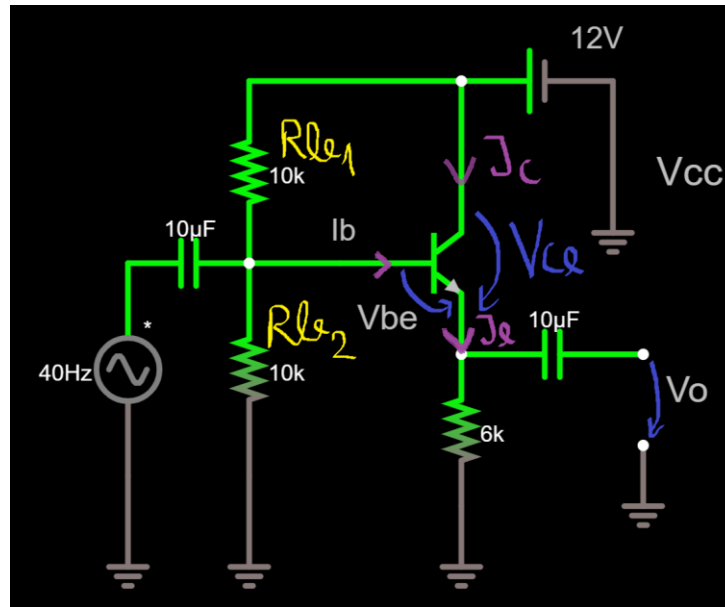
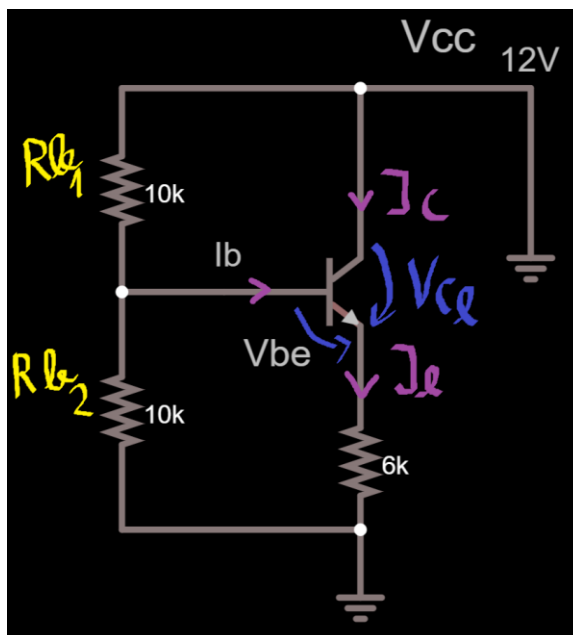


CONEXIUNEA COLECTOR COMUN



În regim de **curent continuu**, avem următoarea schemă:



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(Condensatorul devine întrerupere.)

- În baza tranzistorului avem un divizor de tensiune astfel încât:

$$V_{th} = (R_{b2} / (R_{b1} + R_{b2})) * V_{cc};$$

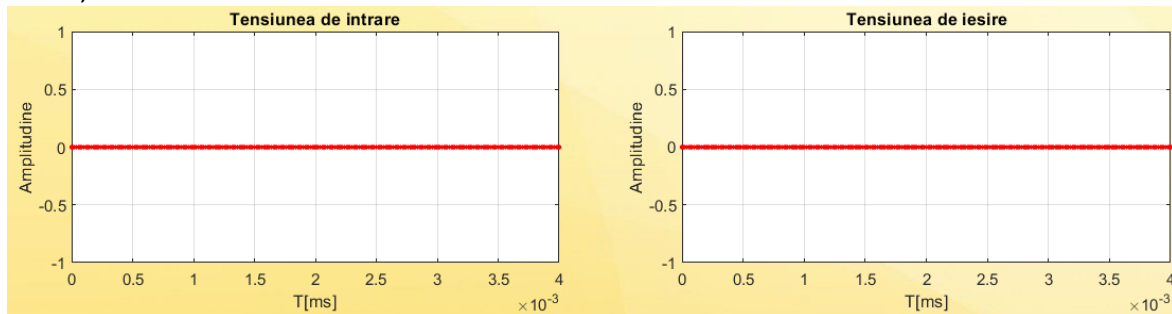
$$R_{th} = (R_{b1} * R_{b2}) / (R_{b1} + R_{b2});$$

Curenții:

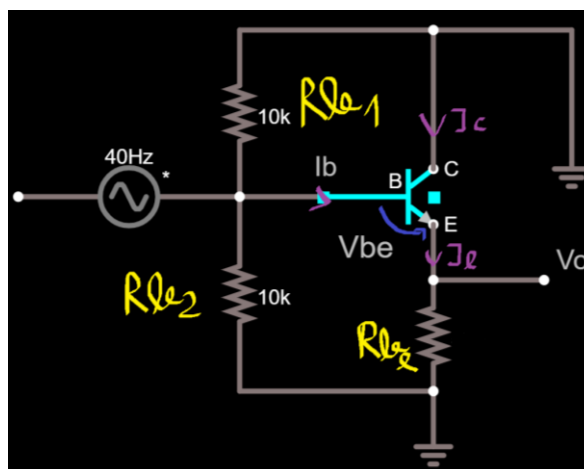
- $I_e = (\beta + 1) * I_b$;
- $I_e = I_c$;
- $I_b = (V_{th} - V_{be}) / (R_{th} + (\beta + 1) * R_e)$;

$v_i = C$; (Componenta continuă)

$v_o = v_i$;



În regim de **curent alternativ**, avem următoarea schemă:



(Condensatorul devine scurtcircuit)

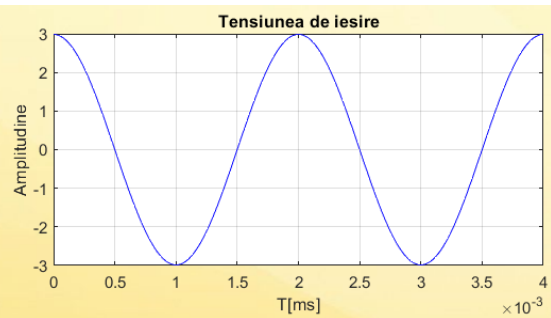
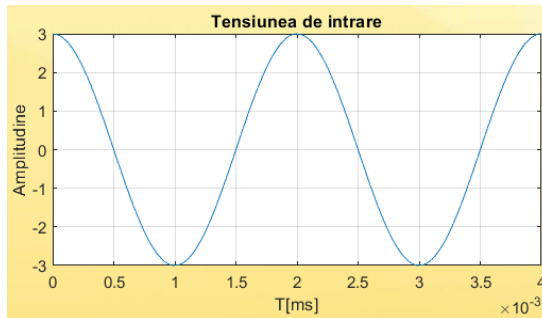
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Tipurile de semnal :

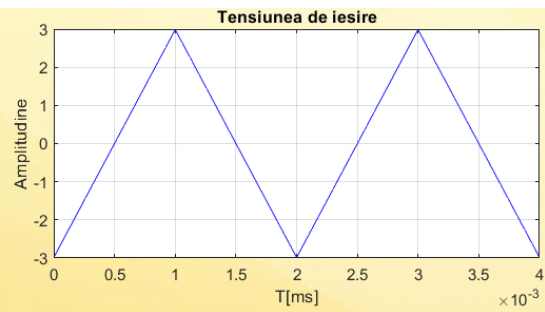
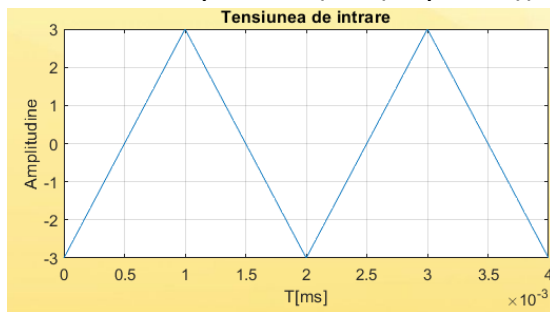
1) Sinusoidal

$$v_i = C + A \cdot \cos(2 \cdot \pi \cdot f \cdot t)$$



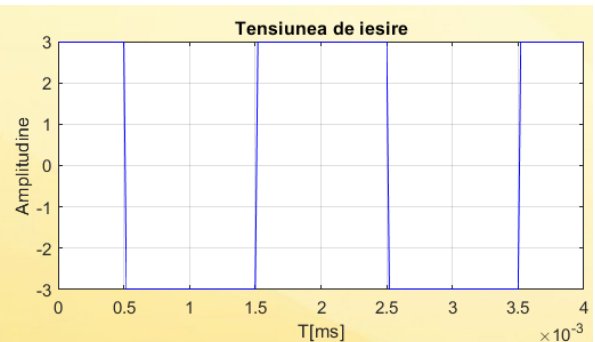
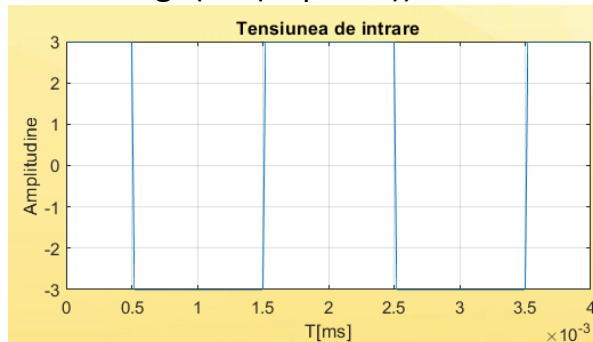
2) Triunghiular:

$$v_i = C + \frac{2 \cdot A}{\pi} \cdot \arccos(\cos(2 \cdot \pi \cdot f \cdot t)) - A$$



3) Dreptunghiular:

$$v_i = C + A \cdot \text{sign}(\cos(2 \cdot \pi \cdot f \cdot t))$$

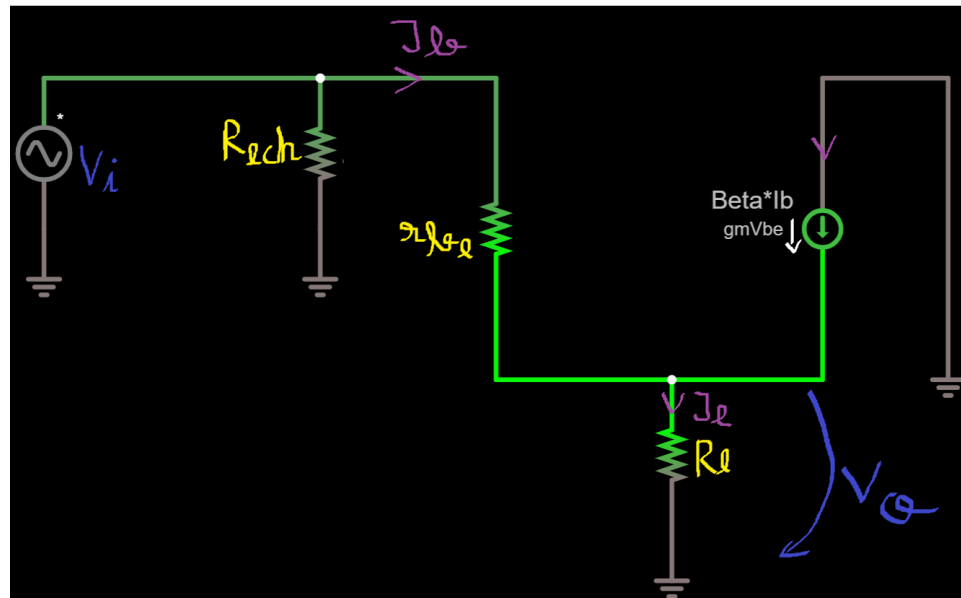


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Schema de semnal mic:

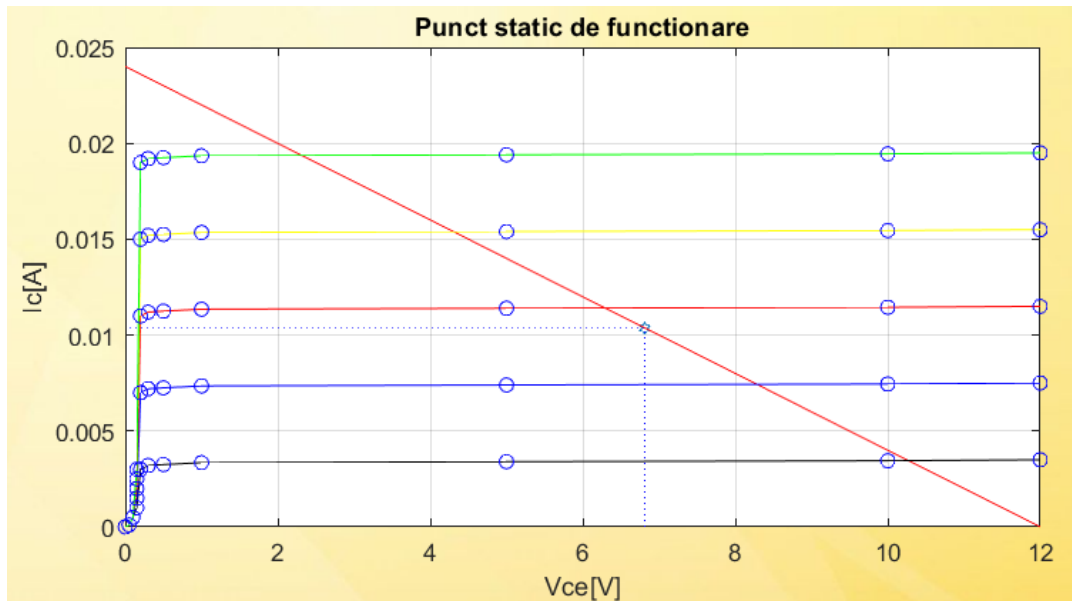
- C»scurt-circuit (sursa de alimentare cu current continuu)
- Colectorul este legat la masa



Parametrii de ieșire:

- ❖ $g_m = 40 \cdot I_c$; (transconductanta)
- ❖ $r_{be} = \beta / g_m$;
- ❖ $R_i = R_e \cdot (r_{be} + (\beta + 1) \cdot R_e) / (R_{th} + r_{be} + (\beta + 1) \cdot R_e)$;
- ❖ $R_0 = R_e \cdot (1 / g_m) / (R_e + 1 / g_m)$; (v_i se pasivizeaza)
- ❖ $A_v = g_m \cdot R_e / (1 + g_m \cdot R_e)$;
- ❖ $V_o = A_v \cdot v_i$

Punctul static de funcționare



$$Q(I_c, V_{ce})$$

$$V_{ce} = V_{cc} - I_e \cdot R_e$$

$$I_c = (\beta + 1) \cdot I_b$$

$$V_{ceSAT} = 0,2V$$

In concluzie:

- Circuitul este un repetor de tensiune (buffer) (Amplitudinea aprox 1) **(dezavantaj)**.
- Rezistența de intrare este foarte mare **(+++)**.
- Rezistența de ieșire este foarte mică **(+++)**.

R0 =	2.40328
Ri =	489.434

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